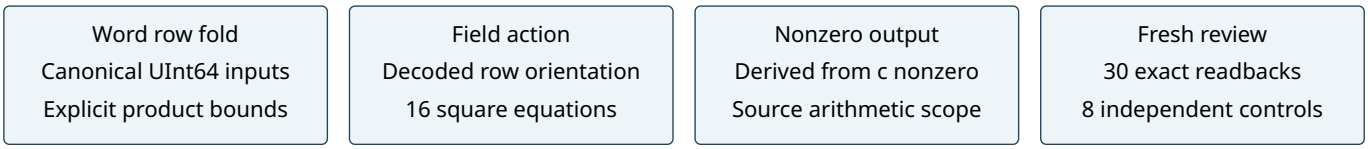
Independent literal Kummer row-fold review

Fresh Lean replay and a native Rust inventory checker passed nine modules and 30 exact named axiom readbacks: 12 main declarations, 10 producer controls and 8 independent controls. The deliberately false theorem was rejected with native exit 1 and a diagnostic proving False is false. No matrix enumeration, point generation or scientific workload ran.



Source fidelity. The bound walk source evaluates four left-fold steps as add(accumulator, mul(matrix[row][column], input[column])). The Lean definitions use that order and the same unsigned modular operations. Canonical operands and p at most 2^32 bound each product below 2^64 and each modular accumulation below 2^64. Four unreduced products sum below 2^66 and therefore fit a 128-bit accumulator. Decoding and nonzero comparison are connected to the existing field and comparator proofs. Counter increments and compiler behavior remain separate.

Square-scalar bridge. The final theorem assumes the sixteen explicit dot-product remainder equations, canonical matrix/input entries, prime p and a canonical nonzero scalar c. It derives the field square-scalar identity and then nonzero preservation of the actual word fold. It does not assume nonzero output. This supplies the arithmetic interface to the earlier matrix certificates; this review does not instantiate all 105 historical records as Lean terms or prove the geometric interpretation of those actions.

Independent controls. A mixed signed block matrix modulo 19 has square 7I, checked over all sixteen entries. A fixed vector checks row orientation, repeated action and application of the final theorem. Other controls distinguish a nonzero word representing field zero, demonstrate multiplication disagreement without canonical entries, check a near-boundary product, and show why the unreduced dot needs more than 64 bits. These are fixed arithmetic controls, not sampled scientific results.

Verification boundary. The producer audit checked the number of unique allowed-axiom declarations. This review additionally checks an explicit ordered inventory of all 30 names and rejects forbidden axioms, an empty inventory, a missing name and a wrong name. Fresh replay includes Product and eight other modules. Local top-level helper sources/binaries are hashed around the final replay; the installed Lean/Mathlib dependency caches retain their existing trust boundary. No independent kernel/binary rechecker was used.

Independence and failures. Root did not author the producer, but has prior familiarity with matrix verification and arithmetic interfaces. The producer run/audit wrappers were read before publication; mathematical source review and fresh verification followed publication at b4684a67ca626d338099525aefae817a3d276e78. No blind re-derivation is claimed. The first independent control failed to synthesize a Decidable canonicality predicate; an explicit finite-index proof corrected it without changing producer source. An initial git source lookup used the wrong working directory and was repeated with the exact worktree. All proof-attempt outputs are retained.